



Assignment 1

Objective:

- Use a push button to toggle a DC motor on and off by utilizing an external interrupt (EXTI) pin on the microcontroller.

Components:

- ATmega32 microcontroller
- DC motor
- Push button
- Resistors
- wires

Circuit Setup:

- Connect the DC motor to the motor driver, and then to the ATmega32 for control.
- Connect the push button to an EXTI pin on the ATmega32 and ground it.

Programming:

- Initialize the EXTI pin for external interrupt detection on the falling edge (button press).
- Configure the microcontroller to toggle the motor state (on/off) each time the interrupt is triggered.
- Ensure proper motor control using PWM signals if necessary for speed regulation.